
Quantum neural operators with implicit quadratic frame and expressivity advantages

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1 Proofs of Quantum Universal Approximation Theorem

In this appendix, we provide rigorous proofs for the approximation capabilities of the proposed quantum neural operator architectures.

1.1 Mathematical Preliminaries

Our derivation begins with the fundamental observation—central to our prior work [1]—that the inner product structure required by classical operator theories can be geometrically embedded into the Hilbert space of quantum states.

Theorem S1 (Embedding of Vector Inner Products into Quantum States) *Let $D_1 \subset \mathbb{R}^{d_1}$ and $D_2 \subset \mathbb{R}^{d_2}$ be compact domains. Let $K_1 \subset C(D_1)$ be a compact subset of the continuous function space. Suppose that $\mathbf{b} : K_1 \rightarrow \mathbb{R}^p$ is a continuous vector-valued functional and $\mathbf{t} : D_2 \rightarrow \mathbb{R}^p$ is a continuous vector-valued function.*

Provided that $\mathbf{b}$ and $\mathbf{t}$ have non-trivial norms, there exists a scaling constant $\lambda \in \mathbb{R}$ and normalized vectors $|b(u)\rangle, |t(\mathbf{y})\rangle \in \mathbb{C}^{p+2}$ such that for all $u \in K_1$ and $\mathbf{y} \in D_2$:

$$\mathbf{b}(u) \cdot \mathbf{t}(\mathbf{y}) = \lambda \langle b(u) | t(\mathbf{y}) \rangle. \quad (1)$$

Let $D_1 \subset \mathbb{R}^{d_1}$ and $D_2 \subset \mathbb{R}^{d_2}$ be compact spatial domains. Let $K_1 \subset C(D_1)$ be a compact set of input functions. The target operator is a continuous mapping $\mathcal{G} : K_1 \rightarrow C(D_2)$.

We invoke the classical universal approximation theorem for nonlinear operators, established by [2], which serves as the foundation for our quantum extensions.

Theorem S2 (Universal Approximation of DeepONet) *Let K_1 be a compact subset of a Banach space of input functions, $D_2 \subset \mathbb{R}^{d_2}$ be a compact domain, and $\mathcal{G} : K_1 \rightarrow C(D_2)$ be a continuous nonlinear operator. For any $\epsilon > 0$, there exists a Deep Operator Network $\mathcal{D}$ parameterized by a latent dimension p such that:*

$$\sup_{u \in K_1, \mathbf{y} \in D_2} \left| \mathcal{G}(u)(\mathbf{y}) - \sum_{k=1}^p b_k(u) t_k(\mathbf{y}) \right| < \epsilon. \quad (2)$$

This theorem implies that any continuous operator can be approximated by a finite dot product of a vector-valued functional $\mathbf{b}(u)$ and a vector-valued function $\mathbf{t}(\mathbf{y})$ in an Euclidean space $\mathbb{R}^p$.

1.2 Proof of Theorem S1

Here we demonstrate that the Euclidean dot product structure can be faithfully embedded into the inner product of normalized quantum states.

Proof First, we define the scaling factors based on the supremum of the Euclidean norms over the compact domains:

$$B := \sup_{u \in K_1} \|\mathbf{b}(u)\|_2, \quad T := \sup_{\mathbf{y} \in D_2} \|\mathbf{t}(\mathbf{y})\|_2.$$

Since K_1 and D_2 are compact and the mappings are continuous, these suprema are finite. By the assumption of non-trivial norms, we have $B > 0$ and $T > 0$. Let the global scaling constant be $\lambda = B \cdot T$.

We define the normalized, dimensionless coefficient functions for $k = 1, \dots, p$:

$$\tilde{b}_k(u) = \frac{b_k(u)}{B}, \quad \tilde{t}_k(\mathbf{y}) = \frac{t_k(\mathbf{y})}{T}.$$

From the definition of B , it follows that $\sum_{k=1}^p |\tilde{b}_k(u)|^2 = \frac{1}{B^2} \|\mathbf{b}(u)\|_2^2 \leq 1$ for all $u \in K_1$ (and similarly for $\tilde{t}_k$ on D_2). Consequently, the terms $1 - \sum_{k=1}^p |\tilde{b}_k(u)|^2$ are non-negative, ensuring that the auxiliary amplitudes defined below are real numbers.

We construct the quantum states in a Hilbert space of dimension $p+2$ ($\cong \mathbb{C}^{p+2}$), spanned by the orthonormal computational basis $\{|1\rangle, \dots, |p\rangle, |p+1\rangle, |p+2\rangle\}$:

$$\begin{aligned} |b(u)\rangle &= \sum_{k=1}^p \tilde{b}_k(u) |k\rangle + \sqrt{1 - \sum_{k=1}^p |\tilde{b}_k(u)|^2} |p+1\rangle, \\ |t(\mathbf{y})\rangle &= \sum_{k=1}^p \tilde{t}_k(\mathbf{y}) |k\rangle + \sqrt{1 - \sum_{k=1}^p |\tilde{t}_k(\mathbf{y})|^2} |p+2\rangle. \end{aligned}$$

These vectors are explicitly normalized (i.e., $\langle b(u)|b(u)\rangle = 1$). Computing their inner product, and utilizing the orthogonality of the basis states ($\langle i|j\rangle = \delta_{ij}$):

$$\begin{aligned} \langle b(u)|t(\mathbf{y})\rangle &= \sum_{k=1}^p \tilde{b}_k(u) \tilde{t}_k(\mathbf{y}) \\ &= \sum_{k=1}^p \frac{b_k(u) t_k(\mathbf{y})}{BT} = \frac{1}{\lambda} \mathbf{b}(u) \cdot \mathbf{t}(\mathbf{y}). \end{aligned}$$

Note that the cross-terms involving $|p+1\rangle$ and $|p+2\rangle$ vanish because $\langle p+1|p+2\rangle = 0$. Rearranging the terms yields the desired identity: $\mathbf{b}(u) \cdot \mathbf{t}(\mathbf{y}) = \lambda \langle b(u)|t(\mathbf{y})\rangle$. Since $\mathbf{b}(u)$ and $\mathbf{t}(\mathbf{y})$ are continuous and B, T are non-zero constants, the composed mappings remain continuous. $\square$

1.3 Proof of Theorem 1

We present a constructive proof of the universal approximation property of QuanONet.

Proof Let $\mathcal{G} : K_1 \rightarrow C(D_2)$ be a continuous operator, with K_1 and D_2 compact. By Theorem S2, for any $\epsilon > 0$ there exist continuous functionals $\hat{b}_k : K_1 \rightarrow \mathbb{R}$ and continuous functions $\hat{t}_k : D_2 \rightarrow \mathbb{R}$, $k = 1, \dots, N$, such that

$$\sup_{u \in K_1, \mathbf{y} \in D_2} \left| \mathcal{G}(u)(\mathbf{y}) - \sum_{k=1}^N \hat{b}_k(u) \hat{t}_k(\mathbf{y}) \right| < \epsilon.$$

Our goal is to construct $\hat{b}_k$ and $\hat{t}_k$ in QuanONet. Since $\hat{b}_k$ and $\hat{t}_k$ are continuous on compact sets, their images are compact and hence bounded. Denote

$$B_{\max} = \max_{1 \leq k \leq N} \sup_{u \in K_1} |\hat{b}_k(u)|, \quad T_{\max} = \max_{1 \leq k \leq N} \sup_{\mathbf{y} \in D_2} |\hat{t}_k(\mathbf{y})|.$$

Choose an integer $p \geq N + 2$ as the dimension of the quantum system. Define scaled amplitudes

$$s_k(\mathbf{y}) = \frac{\hat{t}_k(\mathbf{y})}{\sqrt{p} T_{\max}}, \quad k = 1, \dots, N,$$

and construct a normalized real vector $\mathbf{t}(\mathbf{y}) \in \mathbb{R}^p$ by

$$\begin{aligned} t_1(\mathbf{y}) &= \frac{1}{\sqrt{p}}, \\ t_{k+1}(\mathbf{y}) &= s_k(\mathbf{y}), \quad k = 1, \dots, N, \\ t_{N+2}(\mathbf{y}) &= \sqrt{1 - \frac{1}{p} - \sum_{k=1}^N s_k(\mathbf{y})^2}, \\ t_k(\mathbf{y}) &= 0, \quad k > N + 2. \end{aligned}$$

Then $\|\mathbf{t}(\mathbf{y})\|_2 = 1$ for every $\mathbf{y} \in D_2$, and we set $\mathbf{D}(\mathbf{y}) = \mathbf{t}(\mathbf{y})\mathbf{t}(\mathbf{y})^\top$ since $\mathbf{t}(\mathbf{y})$ is constructed as a real vector, $\mathbf{t}^\top = \mathbf{t}^\dagger$.

Let $\lambda > 0$ and consider the real symmetric arrowhead matrix $\mathbf{M}_0$ defined by

$$(\mathbf{M}_0)_{ii} = 0, \quad (\mathbf{M}_0)_{1,k+1} = (\mathbf{M}_0)_{k+1,1} = \frac{\lambda}{\sqrt{N}} =: r \quad (k = 1, \dots, N),$$

while $(\mathbf{M}_0)_{1,N+2} = (\mathbf{M}_0)_{N+2,1} = 0$ and all other entries are zero. One checks that $\mathbf{M}_0$ is a normal matrix with eigenvalues $\{\lambda, -\lambda, 0, \dots, 0\}$. Hence there exists a unitary $\mathbf{U}_0$ such that $\mathbf{M}_0 = \mathbf{U}_0 \text{diag}[\lambda, -\lambda, 0, \dots, 0] \mathbf{U}_0^\dagger$.

Introduce the diagonal unitary $\psi(u) = \text{diag}[1, e^{i\psi_1(u)}, \dots, e^{i\psi_N(u)}, 1, \dots, 1]$ with phases $\psi_k(u) \in \mathbb{R}$, and define the modulated observable $\mathbf{M}(u) = \psi(u)\mathbf{M}_0\psi(u)^\dagger$. Its nonzero off-diagonal entries are

$$M_{1,k+1}(u) = r e^{-i\psi_k(u)}, \quad M_{k+1,1}(u) = r e^{i\psi_k(u)}, \quad k = 1, \dots, N.$$

Now compute the quantum output:

$$\begin{aligned} \mathcal{Q}(u)(\mathbf{y}) &= \langle \mathbf{M}(u), \mathbf{D}(\mathbf{y}) \rangle_F = \sum_{k=1}^N (M_{1,k+1}(u) D_{k+1,1}(\mathbf{y}) + M_{k+1,1}(u) D_{1,k+1}(\mathbf{y})) \\ &= \sum_{k=1}^N 2r \cos(\psi_k(u)) t_1(\mathbf{y}) t_{k+1}(\mathbf{y}) \\ &= \frac{1}{p T_{\max}} \sum_{k=1}^N 2r \cos(\psi_k(u)) \hat{t}_k(\mathbf{y}), \end{aligned}$$

where we used $t_1(\mathbf{y}) = 1/\sqrt{p}$ and $t_{k+1}(\mathbf{y}) = s_k(\mathbf{y})$. Notice that the normalization component $t_{N+2}(\mathbf{y})$ does not appear because its coupling in $\mathbf{M}_0$ is zero.

Set $c_k(u) = \frac{2r \cos(\psi_k(u))}{p T_{\max}}$, so that $\mathcal{Q}(u)(\mathbf{y}) = \sum_{k=1}^N c_k(u) \hat{t}_k(\mathbf{y})$. The coefficient $c_k(u)$ satisfies $|c_k(u)| \leq \frac{2r}{p T_{\max}} = \frac{2\lambda}{p T_{\max} \sqrt{N}}$. Choose λ large enough that

$$\frac{2\lambda}{p T_{\max} \sqrt{N}} \geq B_{\max}, \quad \text{i.e.} \quad \lambda \geq \frac{p T_{\max} \sqrt{N}}{2} B_{\max}.$$

With such a λ , the numbers

$$\tilde{b}_k(u) = \frac{\hat{b}_k(u) p T_{\max} \sqrt{N}}{2\lambda}$$

belong to $[-1, 1]$. Define $\varphi_k(u) = \arccos(\tilde{b}_k(u))$; each φ_k is a continuous function on K_1 .

Since the Branch network can represent arbitrary continuous functions on K_1 , we may take $\psi_k(u) = \varphi_k(u)$. Then

$$c_k(u) = \frac{2r \cos(\varphi_k(u))}{p T_{\max}} = \hat{b}_k(u),$$

and consequently

$$\mathcal{Q}(u)(\mathbf{y}) = \sum_{k=1}^N \hat{b}_k(u) \hat{t}_k(\mathbf{y}).$$

Thus, for any $\epsilon > 0$, by choosing p sufficiently large and λ satisfying the inequality above, QuanONet approximates $\mathcal{G}$ with error less than ϵ . This completes the proof. $\square$

2 Proof of the Implicit Quadratic Advantage

In this appendix, we provide a rigorous mathematical foundation for the implicit quadratic advantage of QuanONet.

2.1 Mathematical Preliminaries

We first establish the strict equivalence between minimizing the reconstruction error of the network and maximizing the capture of the dominant eigenspace of the covariance operator. Assuming without loss of generality that the target operator outputs are centered (i.e., $\mathbb{E}_{v \sim \mathcal{G}_{\#}\mu}[v] = 0$), we define the covariance operator as follows:

Definition S1 (Covariance Operator) Let μ be the probability measure of the input space, and $\mathcal{G}_{\#}\mu$ be the pushforward measure on the output function space $L^2(D_2)$. The covariance operator $\Gamma_{\mathcal{G}_{\#}\mu} : L^2(D_2) \rightarrow L^2(D_2)$ is defined such that for any function $w \in L^2(D_2)$,

$$\Gamma_{\mathcal{G}_{\#}\mu} w = \int_{L^2(D_2)} \langle w, v \rangle v \, d(\mathcal{G}_{\#}\mu)(v), \quad (3)$$

where $v = \mathcal{G}(u)$ represents the output random element.

We note that assuming the pushforward measure $\mathcal{G}_{\#}\mu$ has a finite second moment (i.e., $\mathbb{E}_{v \sim \mathcal{G}_{\#}\mu}[\|v\|_{L^2(D_2)}^2] < \infty$), the covariance operator $\Gamma_{\mathcal{G}_{\#}\mu}$ is naturally self-adjoint and positive semi-definite due to the symmetry and linearity of the inner product. Furthermore, the finite second moment guarantees that the operator has a finite trace. Being a positive semi-definite trace-class operator, $\Gamma_{\mathcal{G}_{\#}\mu}$ is inherently compact. Consequently, there exists a sequence of non-negative eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq 0$ with $\sum_{i=1}^{\infty} \lambda_i < \infty$ and a corresponding orthonormal basis of eigenfunctions $\{\phi_i\}_{i=1}^{\infty}$.

For a linear space V spanned by the network's basis functions, the reconstruction error $\hat{\mathcal{E}}_{\mathcal{R}}(V)$ is defined as the mean squared projection error, which can be formulated via the trace of the covariance operator.

For a width- p DeepONet, all possible outputs lie in

$$V_{\mathcal{D}} = \text{span}\{t_1, \dots, t_p\} \subset L^2(D_2), \quad \dim V_{\mathcal{D}} \leq p.$$

QuanONet uses a width- p trunk vector $\mathbf{t}(\mathbf{y}) = [t_1(\mathbf{y}), \dots, t_p(\mathbf{y})]^\top$. Let $\mathbf{D}(\mathbf{y}) = \mathbf{t}(\mathbf{y})\mathbf{t}(\mathbf{y})^\top$. Since D_2 is compact and each t_i is continuous (hence bounded), we have $t_i t_j^* \in L^2(D_2)$. All possible outputs lie in

$$V_{\mathcal{D}} = \text{span}\{\Re(t_i t_j^*), \Im(t_i t_j^*) : 1 \leq i, j \leq p\} \subset L^2(D_2), \quad \dim V_{\mathcal{D}} \leq p^2.$$

The quadratic count is attained when the product functions are linearly independent.

Let P_V be the orthogonal projection onto V . The error is exactly the residual trace:

$$\hat{\mathcal{E}}_{\mathcal{D}}(V) = \mathbb{E}_{u \sim \mu} \left[\|\mathcal{G}(u) - P_V \mathcal{G}(u)\|_{L^2(D_2)}^2 \right] = \text{Tr}(\Gamma_{\mathcal{G}_{\#}\mu}) - \text{Tr}(P_V \Gamma_{\mathcal{G}_{\#}\mu}). \quad (4)$$

Minimizing $\hat{\mathcal{E}}_{\mathcal{D}}(V)$ is therefore strictly equivalent to maximizing the captured variance $\text{Tr}(P_V \Gamma_{\mathcal{G}_{\#}\mu})$.

Theorem S3 (Ky Fan Theorem) *For any compact self-adjoint operator Γ and any rank- r orthogonal projection P , we have:*

$$\max_{\text{rank}(P_V)=r} \text{Tr}(P_V \Gamma) = \sum_{i=1}^r \lambda_i, \quad (5)$$

where the maximum is attained by the subspace $V_r^{\text{opt}} = \text{span}\{\phi_1, \dots, \phi_r\}$ [3, 4].

2.2 Core Concepts in Additive Combinatorics

To analyze the expressive power of QuanONet in the frequency domain, we introduce necessary tools from additive combinatorics.

Definition S2 (Sumsets and Difference Sets) For any finite subset of d -dimensional integer lattice $A \subset \mathbb{Z}^d$, its sumset $A + A$ and difference set $A - A$ are defined as:

$$A + A = \{\mathbf{a} + \mathbf{b} \mid \mathbf{a}, \mathbf{b} \in A\}, \quad (6)$$

$$A - A = \{\mathbf{a} - \mathbf{b} \mid \mathbf{a}, \mathbf{b} \in A\}. \quad (7)$$

Definition S3 (Small Doubling Constant) A finite set $A \subset \mathbb{Z}^d$ is said to have a small doubling constant $K \geq 1$ if $|A + A| \leq K|A|$.

Definition S4 (Sidon Set) A set $S \subset \mathbb{Z}^d$ is a Sidon set if all pairwise sums of its elements are distinct. That is, $\mathbf{a} + \mathbf{b} = \mathbf{c} + \mathbf{d}$ for $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d} \in S$ implies $\{\mathbf{a}, \mathbf{b}\} = \{\mathbf{c}, \mathbf{d}\}$.

Remark 1 (Examples of Additive Structure) The doubling constant characterizes the additive algebraic structure of a set. Most highly structured sets naturally possess a small doubling constant. For instance, intuitively in a 1D scenario, if $A = \{1, 2, \dots, n\}$ is a standard arithmetic progression, its sumset is $A + A = \{2, 3, \dots, 2n\}$, yielding $|A + A| = 2n - 1 < 2|A|$. Thus, its doubling constant is bounded by $K = 2$.

Conversely, a Sidon set represents extreme additive un-structure, maximizing the size of the sumset to $|S + S| = \frac{|S|(|S|+1)}{2} = \mathcal{O}(|S|^2)$. Such sets are extremely sparse and "rare" in dense intervals. A classic 1D example of a Sidon set is the sequence of powers of 2, $S = \{1, 2, 4, 8, \dots, 2^m\}$, where every pairwise sum is strictly unique due to the uniqueness of binary representations. In our context, Sidon sets serve as the extreme theoretical worst-case where the implicit quadratic advantage completely fails.

Theorem S4 (Freiman-Ruzsa Theorem) *If $A \subset \mathbb{Z}^d$ satisfies $|A + A| \leq K|A|$, then A is contained in a Generalized Arithmetic Progression (GAP) P of algebraic dimension $d_{GAP}(K)$ and size $|P| \leq f(K)|A|$, where $d_{GAP}(K)$ and $f(K)$ depend on K .*

2.3 Spectral Equivalence and the Failure Case

We set the output space to be the d -dimensional torus $D_2 = \mathbb{T}^d$. We assume that the optimal eigenfunctions of the covariance operator are Fourier modes. To establish a rigorous discrete mapping, we correspondingly assume that the basis functions learned by the QuanONet trunk state are pure Fourier modes. While this restriction is not fully general and theoretically weakens the trunk's actual non-linear expressive power, it provides a strictly conservative lower bound for our analysis.

Let $\Xi \subset \mathbb{Z}^d$ be the frequency index set corresponding to the top p^2 eigenvalues of the covariance operator, i.e., $|\Xi| = p^2$. Let $\Omega \subset \mathbb{Z}^d$ be the set of frequencies sampled by the QuanONet trunk state of dimension p , with $|\Omega| = p$. Under the complex exponential basis, the pointwise multiplication of trunk outputs (with complex conjugation) maps to the subtraction of their frequency indices. Consequently, the quadratic feature space $V_{\mathcal{Q}}$ of QuanONet corresponds to the difference set $(\Omega - \Omega)$.

By Theorem S3, maximizing the spectral capture $\text{Tr}(P_{V_{\mathcal{Q}}} \Gamma)$ relies on successfully overlapping the model's feature space with the dominant eigen-frequencies. To isolate the underlying structural capability of the network, we abstract this continuous objective into a purely combinatorial frequency-matching problem:

$$\max_{|\Omega|=p} |(\Omega - \Omega) \cap \Xi|. \quad (8)$$

The implicit quadratic advantage derived from the difference set is not unconditional. If the target spectrum Ξ is structurally dissociated, the representational advantage degenerates.

Lemma S1 (Expressivity Degradation on Sidon Sets) *Assume the target dominant spectrum Ξ (where $|\Xi| = p^2$) forms a Sidon set. Then for any latent frequency set Ω of size p , the effective spectral coverage is strictly bounded by $|(\Omega - \Omega) \cap \Xi| \leq \mathcal{O}(p^{3/2})$.*

Proof Consider a bipartite graph $G = (\Omega, \Omega, E)$ where an edge $(\mathbf{u}, \mathbf{v})$ exists if and only if their difference $\mathbf{u} - \mathbf{v} \in \Xi$. Suppose there exists a cycle of length 4 (C_4) with vertices $\mathbf{u}_1, \mathbf{v}_1, \mathbf{u}_2, \mathbf{v}_2$. The alternating sum of the edge weights inherently yields $(\mathbf{u}_1 - \mathbf{v}_1) - (\mathbf{u}_2 - \mathbf{v}_1) + (\mathbf{u}_2 - \mathbf{v}_2) - (\mathbf{u}_1 - \mathbf{v}_2) = \mathbf{0}$, implying $\lambda_1 + \lambda_3 = \lambda_2 + \lambda_4$ for some $\lambda_i \in \Xi$.

However, since Ξ is a Sidon set, the equation $\mathbf{a} + \mathbf{b} = \mathbf{c} + \mathbf{d}$ permits only trivial solutions where $\{\mathbf{a}, \mathbf{b}\} = \{\mathbf{c}, \mathbf{d}\}$. This combinatorial rigidity guarantees that the graph G is entirely C_4 -free. By the Kővári–Sós–Turán theorem, the maximum number of edges in a C_4 -free bipartite graph with parts of size p is strictly bounded by $\mathcal{O}(p^{3/2})$. Thus, the maximum intersection size satisfies $|(\Omega - \Omega) \cap \Xi| \leq \mathcal{O}(p^{3/2}) \ll p^2$. The effective spectral dimension is consequently highly degenerate. $\square$

2.4 Proof of Theorem 2

To rigorously establish the expressivity bounds in Theorem 2, we first prove a foundational combinatorial lemma regarding the spectral coverage of the difference set.

Lemma S2 If $p \geq 3$ and the target spectrum Ξ ($|\Xi| = p^2$) satisfies $|\Xi + \Xi| \leq K|\Xi|$, then there exists a subset $\Omega \subset \mathbb{Z}^d$ with $|\Omega| = p$ and a constant $\kappa > 0$ independent of p , such that:

$$|(\Omega - \Omega) \cap \Xi| \geq \kappa p^2. \quad (9)$$

Proof Step 1: GAP Embedding. By the Freiman-Ruzsa theorem (Theorem S4), the small doubling condition $|\Xi + \Xi| \leq K|\Xi|$ implies that Ξ is structurally constrained. Specifically, Ξ is contained within a proper Generalized Arithmetic Progression (GAP) P of dimension $d_{\text{GAP}} \leq d_{\text{GAP}}(K)$ and bounded volume $|P| \leq f(K)|\Xi| = f(K)p^2$, where $f(K)$ is a constant depending only on K .

Step 2: Expansion and Representation. To successfully capture arbitrary frequencies (even strictly high-frequency bands not centered at the origin), we define an extended pool $P_{\text{ext}} = P \cup (P + P)$. A fundamental property of GAPs guarantees that its volume scales boundedly: $|P_{\text{ext}}| \leq |P| + |P + P| \leq (1 + 2^{d_{\text{GAP}}})|P| \leq (1 + 2^{d_{\text{GAP}}(K)})f(K)p^2 = C_K p^2$, where C_K is a combined constant depending only on K .

Crucially, for any target frequency $\mathbf{x} \in \Xi \subset P$, we can construct differences yielding $\mathbf{x}$ by fixing $\mathbf{b} \in P$ and setting $\mathbf{a} = \mathbf{x} + \mathbf{b}$. By definition, $\mathbf{a} \in P + P$. Thus, both $\mathbf{a}$ and $\mathbf{b}$ belong to P_{ext} , and their difference is rigorously $\mathbf{a} - \mathbf{b} = \mathbf{x}$. Since there are $|P|$ choices for $\mathbf{b}$, there are exactly $|P|$ such pairs $(\mathbf{a}, \mathbf{b})$ in $P_{\text{ext}} \times P_{\text{ext}}$. Treating these pairs as a directed graph where each vertex has an in-degree and out-degree of at most 1, we can extract a set of at least $\nu \geq \lfloor |P|/3 \rfloor$ disjoint pairs in $P_{\text{ext}} \times P_{\text{ext}}$ whose difference strictly yields $\mathbf{x}$.

Step 3: Random Sampling. We construct a random subset $S \subset P_{\text{ext}}$ by including each element of P_{ext} independently with probability $\rho = \frac{p}{|P_{\text{ext}}|}$. By linearity of expectation, the expected size of this random set is $\mathbb{E}[|S|] = \rho|P_{\text{ext}}| = p$. For a fixed $\mathbf{x} \in \Xi$, assuming p is sufficiently large such that the disjoint pairs $\nu \geq \lfloor |P|/3 \rfloor \geq |P|/4$, the probability that no pair in S yields the difference $\mathbf{x}$ is bounded by the probability of failing on all ν disjoint pairs:

$$\mathbb{P}(\mathbf{x} \notin S - S) \leq (1 - \rho^2)^\nu \leq \exp(-\rho^2 \nu) \leq \exp\left(-\frac{p^2}{|P_{\text{ext}}|^2} \frac{|P|}{4}\right). \quad (10)$$

Using the properties $|P_{\text{ext}}| \leq (1 + 2^{d_{\text{GAP}}})|P|$ and $|P| \leq f(K)p^2$, we can bound the exponent strictly away from zero:

$$\frac{p^2|P|}{|P_{\text{ext}}|^2} \geq \frac{p^2|P|}{(1 + 2^{d_{\text{GAP}}})^2|P|^2} = \frac{p^2}{(1 + 2^{d_{\text{GAP}}})^2|P|} \geq \frac{1}{(1 + 2^{d_{\text{GAP}}})^2 f(K)}. \quad (11)$$

Thus, $\mathbb{P}(\mathbf{x} \notin S - S) \leq \exp\left(-\frac{1}{4(1+2^{d_{\text{GAP}}})^2 f(K)}\right)$. This failure probability is strictly bounded by $1 - \delta$ for $\delta = 1 - \exp\left(-\frac{1}{4(1+2^{d_{\text{GAP}}})^2 f(K)}\right) > 0$. By linearity of expectation over all $\mathbf{x} \in \Xi$, the expected size of the intersection is $\mathbb{E}[(S - S) \cap \Xi] \geq \delta p^2$.

Step 4: Alteration Method. We must convert the random set S (expected size p) into a deterministic set Ω of exact size p . To control the size of S while preserving the intersection, we introduce the random variable $Z = |(S - S) \cap \Xi| - \frac{\delta}{2}|S|^2$. Utilizing the bound on the second moment $\mathbb{E}[|S|^2] = \text{Var}(|S|) + (\mathbb{E}[|S|])^2 \leq p + p^2$, its expectation is:

$$\mathbb{E}[Z] \geq \delta p^2 - \frac{\delta}{2}(p^2 + p) = \frac{\delta}{2}p^2 - \frac{\delta}{2}p. \quad (12)$$

For $p \geq 2$, $\mathbb{E}[Z] \geq \frac{\delta}{4}p^2$. Thus, there exists a deterministic choice of $S_0 \subset P_{\text{ext}}$ such that $|(S_0 - S_0) \cap \Xi| - \frac{\delta}{2}|S_0|^2 \geq \frac{\delta}{4}p^2$. This guarantees that $|S_0| \leq \sqrt{2/\delta} \cdot p$, meaning $|S_0| = \mathcal{O}(p)$.

If $|S_0| > p$, we uniformly at random downsample S_0 to exactly p elements, forming the final set Ω . Since a target difference survives if at least one of its generating pairs is preserved, the expected number of surviving differences is scaled by a factor of at least the hypergeometric probability $\frac{p(p-1)}{|S_0|(|S_0|-1)}$. Since $|S_0| = \mathcal{O}(p)$, this scaling factor is bounded by a constant strictly greater than 0. Applying this deterministic downsampling yields $|(\Omega - \Omega) \cap \Xi| \geq \kappa p^2$ for some universal constant $\kappa > 0$ depending only on K , completing the proof. $\square$

Substituting the combinatorial lower bound from Lemma S2 back into the projection error formula, we find that under the small doubling condition, QuanONet requires a latent dimension of only $\mathcal{O}(p)$ to match the subspace capture—and equivalently, the reconstruction bound—that would necessitate a trunk width of $\mathcal{O}(p^2)$ in a standard DeepONet. This completes the rigorous proof of Theorem 2.

The assumption that the target spectrum Ξ exhibits a small doubling constant ($|\Xi + \Xi| \leq K|\Xi|$) is not merely a mathematical convenience; it is naturally satisfied by the eigensystems of most physical processes. Due to the well-known spectral bias of neural networks and the compactness of physical kernel operators (e.g., Green’s functions of PDEs, Gaussian or Matérn kernels), dominant eigenvalues typically cluster densely around low frequencies or form regular grids. Such dense, contiguous frequency clusters strictly possess small doubling constants (often $K \approx 2^d$).

To intuitively illustrate this combinatorial advantage, consider a simplified 1D scenario where the dominant spectrum forms a contiguous block including the zero-mode: $\Xi = \{0, 1, \dots, p^2 - 1\}$. A standard DeepONet would require an explicit trunk width of p^2 to perfectly reconstruct this target space.

For QuanONet, we can explicitly construct a base frequency set Ω as the union of a coarse grid and a fine grid:

$$\Omega = \{p, 2p, 3p, \dots, p^2\} \cup \{0, 1, 2, \dots, p - 1\}. \quad (13)$$

The total dimension cost is $|\Omega| = 2p$. However, the difference set $(\Omega - \Omega)$ inherently contains all cross-differences between the coarse and fine grids. For any coarse element

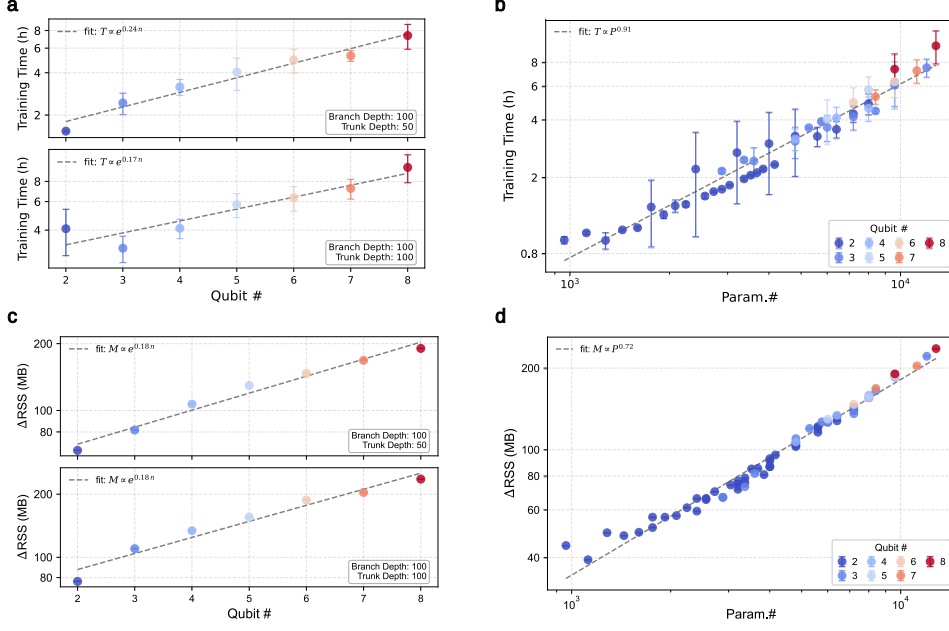


Fig. S1 Empirical scaling behavior of computational overheads. The curves illustrate the training time versus (a) the number of qubits and (b) the number of parameters, alongside the corresponding memory footprint scaling fits versus (c) the number of qubits and (d) the number of parameters. Data are presented as mean values $\pm$ SD derived from $n = 5$ independent trials with randomized network initializations and data splits.

kp ($k \in \{1, 2, \dots, p\}$) and fine element j ($j \in \{0, 1, \dots, p-1\}$), the difference $kp - j$ effortlessly generates every exact integer from 1 to p^2 , while self-differences naturally generate 0. Thus, $(\Omega - \Omega)$ perfectly encapsulates Ξ , yielding $|(\Omega - \Omega) \cap \Xi| = p^2$. With only $\mathcal{O}(p)$ basis functions, QuanONet successfully synthesizes the $\mathcal{O}(p^2)$ -dimensional optimal subspace.

In practical machine learning tasks, Ξ might not perfectly adhere to the strict small doubling condition due to spectral noise or isolated high-frequency modes. However, the implicit quadratic advantage remains highly robust. From the perspective of additive combinatorics, if Ξ has sufficiently large additive energy, the Balog-Szemerédi-Gowers theorem guarantees a substantial subset $\Xi_{\text{core}} \subset \Xi$ with a bounded doubling constant. Since QuanONet efficiently captures this dense structural core Ξ_{core} , the quadratic frequency-coverage mechanism therefore persists on this structured core.

3 Wall-clock Times and Memory Footprints

We report the empirical training time and memory footprint as functions of parameter capacity and qubit count to establish a transparent computational baseline. Under a fixed system size, training time scales almost linearly with parameter count ($T \propto$

$P^{0.91}$, Fig. S1b). Due to intermediate tensor allocations required for backpropagation, memory overhead scales sub-linearly as $M \propto P^{0.72}$ (Fig. S1d).

To assess classical simulation scalability limits, we analyze overhead across varying qubit counts n under fixed circuit depths. Our fits capture the inherent exponential complexity: training time scales as $T \propto e^{0.24n}$ for depths ($h_b = 100, h_t = 50$) and $T \propto e^{0.17n}$ for ($h_b = 100, h_t = 100$) (Fig. S1a). Concurrently, the memory footprint exhibits a parallel exponential trend of $M \propto e^{0.18n}$ (Fig. S1c).

Despite these fundamental limitations, our maximum verification regime ($p = 256$, $n = 8$ qubits) remains highly tractable. The peak memory footprint for the 8-qubit system is only ~ 200 MB, identifying dense tensor contractions—not memory exhaustion—as the primary bottleneck. These scaling laws confirm that while simulator overhead is manageable for near-term validation ($n \leq 8$), bypassing the exponential temporal and spatial walls inherently necessitates native execution on physical quantum hardware.

4 Further Related Works

The landscape of quantum algorithms for solving Partial Differential Equations (PDEs) has evolved rapidly, generally categorized into three distinct paradigms: fault-tolerant quantum simulation, variational quantum algorithms (VQAs), and the emerging field of quantum neural operators.

4.1 Quantum Simulation and Linear Algebra Subroutines

A theoretical foundation for quantum PDE solving was established by the Harrow-Hassidim-Lloyd (HHL) algorithm [5], which proposed a method to solve linear systems of equations $\mathbf{Ax} = \mathbf{b}$ with complexity scaling logarithmically with the system size $\mathcal{O}(\log N)$, offering an exponential speedup over classical conjugate gradient methods. However, HHL requires deep quantum circuits and efficient Quantum Random Access Memory (QRAM), limiting its immediate applicability in the NISQ era. To address general linear PDEs—including non-Hermitian and dissipative systems like the heat or convection-diffusion equations—[6] introduced the Schrödingerization framework. This technique employs a warped phase transformation to embed non-unitary dynamics into higher-dimensional unitary evolution, allowing standard Hamiltonian simulation methods to be applied to a wider class of PDEs. More recently, this approach has been extended to nonlinear equations via the HAM-Schrödingerization method [7], which combines homotopy analysis with quantum simulation to iteratively solve linearized sub-problems.

4.2 Variational Quantum Algorithms (VQAs)

Acknowledging the hardware constraints of the NISQ era, research has shifted towards hybrid quantum-classical approaches. The Variational Quantum Linear Solver (VQLS) [8] replaces the deep phase estimation circuits of HHL with a shallow, parameterized ansatz, optimizing a local cost function to approximate the solution of linear systems. In the context of differential equations, Differentiable Quantum Circuits

(DQC) [9] have emerged as a quantum analogue of PINNs. These methods utilize the parameter-shift rule to analytically compute gradients of quantum circuits, enabling the incorporation of PDE residuals directly into the loss function. While VQAs significantly reduce circuit depth, they face challenges related to trainability, such as the barren plateau problem, where gradients vanish exponentially with system size.

4.3 Detailed Comparison with QDeepONet

To explicitly contextualize the proposed architecture, we outline the primary distinctions between QuanONet and the recently proposed QDeepONet [10]. First, regarding structural design, QuanONet is a native, end-to-end quantum framework in which both feature mapping and nonlinear transformations are implemented on quantum circuits, enabling potential quantum-aware training. In contrast, QDeepONet adopts a hybrid paradigm that utilizes quantum processing primarily for accelerating linear matrix multiplications during inference, while relying on classical processors for nonlinear activation functions and the entire training process. Second, in terms of resource efficiency, QuanONet leverages amplitude embedding principles to compress the latent space, requiring only $\mathcal{O}(\log p)$ qubits to represent a network width of p . Conversely, QDeepONet employs unary encoding, which necessitates a qubit count scaling as $\mathcal{O}(p)$, thereby imposing substantial hardware overhead for wide networks. Finally, the nature of the quantum utility differs significantly. QDeepONet aims for computational speedup during the evaluation pass but does not inherently expand the theoretical approximation capacity relative to its classical counterpart (and may be susceptible to noise-induced degradation). QuanONet, however, exploits the density matrix to implicitly construct a quadratic frame of size $\mathcal{O}(p^2)$ from an $\mathcal{O}(p)$ latent dimension. This provides a provable representational advantage, enabling enhanced approximation accuracy for complex operators, as summarized in Table S1.

Table S1 Structural and functional comparison between QDeepONet and QuanONet. Here p denotes the width of the classical network counterpart.

Feature	QDeepONet [10]	QuanONet (Ours)
Architecture Type	Hybrid	Native Quantum
Qubit Complexity	$\mathcal{O}(p)$	$\mathcal{O}(\log p)$
Training & Inference	Classical Training; Hybrid Inference	Quantum Training & Inference
Frame Construction	Explicit Linear ($\mathcal{O}(p)$)	Implicit Quadratic ($\mathcal{O}(p^2)$)
Primary Advantage	Evaluation Speedup	Representational Capacity

4.4 Theoretical Connection to FNO

Recent theoretical advancements by [11] have established a unifying perspective on operator learning, demonstrating that the FNO can be mathematically interpreted

as a specialized instance of the DeepONet architecture. In this framework, the FNO corresponds to a DeepONet where the Trunk Net is structurally fixed to approximate a trigonometric basis, effectively performing a kernel convolution in the frequency domain. This insight invites a theoretical comparison regarding how frame functions are constructed in each architecture. The standard FNO paradigm explicitly parameterizes the integral operator using the FFT. By operating on a discrete integer lattice $\mathbb{Z}^d$ and applying a frequency truncation determined by a fixed bandwidth parameter, FNO efficiently learns the coefficients of standard harmonic modes e^{iky} . This design acts as a fixed spectral filter, prioritizing low-frequency modes to achieve rapid convergence for operators dominated by smooth dynamics but limiting the basis to a fixed harmonic grid. In contrast, TF-QuanONet approaches spectral representation through a continuous mechanism governed by quantum evolution. Rather than pre-defining a fixed spectral grid, the Trunk layers generate encoding functions via the evolution of a state vector under a parameterized Hamiltonian. A defining feature of this quantum architectural design is the incorporation of the Trainable-Frequency strategy. By treating embedding frequencies as continuous learnable parameters rather than fixed integers, the model retains the flexibility to adjust its spectral alignment during training. This mechanism allows the architecture to represent frequencies that may be non-commensurate or lie off the standard integer lattice. Consequently, while both architectures leverage spectral principles to capture global correlations, they diverge in their basis construction strategy: FNO employs a fixed, discrete trigonometric basis optimized for computational speed via FFT, whereas TF-QuanONet synthesizes an adaptive spectral representation driven by the continuous parameters of the quantum circuit.

Table S2 Comparison of spectral handling between FNO and TF-QuanONet.

Feature	FNO [12]	TF-QuanONet (Ours)
Trunk Architecture	Fixed Trigonometric Basis	Quantum State Evolution
Spectral Domain	Discrete Integer Lattice ($\mathbb{Z}^d$)	Continuous Real Spectrum ($\mathbb{R}^d$)
Frequency Handling	Fixed Modes (Truncated)	Learnable Parameters (Adaptive)
Primary Mechanism	High-frequency Filtering	Spectral Alignment

4.5 Connections with Classical Machine Learning Representations

The implicit quadratic frame construction, utilized here to enhance the expressivity of quantum neural operators, exhibits fundamental structural parallels with established classical machine learning paradigms designed for high-order feature interaction. Mathematically, the utilization of the Frobenius inner product between a modulated observable and a rank-1 density matrix formally implements a homogeneous polynomial kernel of degree two, implicitly mapping latent representations into a tensor product space of size $\mathcal{O}(p^2)$ without explicit coordinate expansion [13]. This

reliance on the outer product structure to aggregate coordinate information establishes a direct methodological correspondence to bilinear pooling techniques in computer vision [14], where the density matrix functions as a global covariance descriptor capturing second-order statistical dependencies among latent state amplitudes. Furthermore, the operational form of the interaction mirrors linear attention mechanisms [15], as the density matrix serves as a compressed context representation that decouples input and output embeddings while preserving dense connectivity via the spectral trace operation. Finally, the introduction of the Trainable-Frequency strategy to resolve spectral dimensional collapse aligns with latent vector optimization in Factorization Machines [16]; specifically, learning continuous embedding parameters ensures the linear independence of pairwise spectral differences, thereby preventing degeneracy and preserving the expanded expressivity of the quadratic frame.

References

- [1] Wang, R., Xia, Z., Yan, G. & Yan, J. *Quanonet: Quantum neural operator with application to differential equation*. International Conference on Machine Learning (2025).
- [2] Chen, T. & Chen, H. Universal approximation to nonlinear operators by neural networks with arbitrary activation functions and its application to dynamical systems. *IEEE transactions on neural networks* **6**, 911–917 (1995).
- [3] Fan, K. On a theorem of weyl concerning eigenvalues of linear transformations i*. *Proceedings of the National Academy of Sciences* **35**, 652–655 (1949). URL <https://www.pnas.org/doi/abs/10.1073/pnas.35.11.652>.
- [4] Fan, K. Maximum properties and inequalities for the eigenvalues of completely continuous operators*. *Proceedings of the National Academy of Sciences* **37**, 760–766 (1951). URL <https://www.pnas.org/doi/abs/10.1073/pnas.37.11.760>.
- [5] Harrow, A. W., Hassidim, A. & Lloyd, S. Quantum algorithm for linear systems of equations. *Physical review letters* **103**, 150502 (2009).
- [6] Jin, S., Liu, N. & Yu, Y. Quantum simulation of partial differential equations via Schrödingerization. *Physical Review Letters* **133**, 230602 (2024).
- [7] Liao, S. Ham-schrödingerisation: a generic framework of quantum simulation for any nonlinear pdes. *arXiv preprint arXiv:2406.15821* (2024).
- [8] Bravo-Prieto, C. *et al.* Variational quantum linear solver. *arXiv preprint arXiv:1909.05820* (2019).
- [9] Kyriienko, O., Paine, A. E. & Elfving, V. E. Solving nonlinear differential equations with differentiable quantum circuits. *Physical Review A* **103**, 052416 (2021).

- [10] Xiao, P., Zheng, M., Jiao, A., Yang, X. & Lu, L. Quantum deeponet: Neural operators accelerated by quantum computing. *arXiv preprint arXiv:2409.15683* (2024).
- [11] Kovachki, N., Lanthaler, S. & Mishra, S. On universal approximation and error bounds for fourier neural operators. *Journal of Machine Learning Research* **22**, 1–76 (2021).
- [12] Li, Z. *et al.* *Fourier neural operator for parametric partial differential equations*. International Conference on Learning Representations (2021).
- [13] Schölkopf, B. & Smola, A. J. *Learning with kernels: support vector machines, regularization, optimization, and beyond* (MIT press, 2002).
- [14] Lin, T.-Y., RoyChowdhury, A. & Maji, S. *Bilinear cnn models for fine-grained visual recognition*. Proceedings of the IEEE international conference on computer vision, 1449–1457 (2015).
- [15] Katharopoulos, A., Vyas, A., Pappas, N. & Fleuret, F. *Transformers are rnns: Fast autoregressive transformers with linear attention*. International conference on machine learning, 5156–5165 (PMLR, 2020).
- [16] Rendle, S. *Factorization machines*. 2010 IEEE International conference on data mining, 995–1000 (IEEE, 2010).